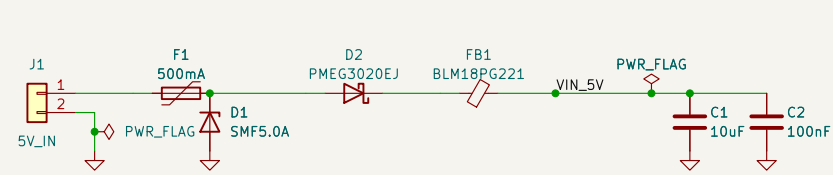
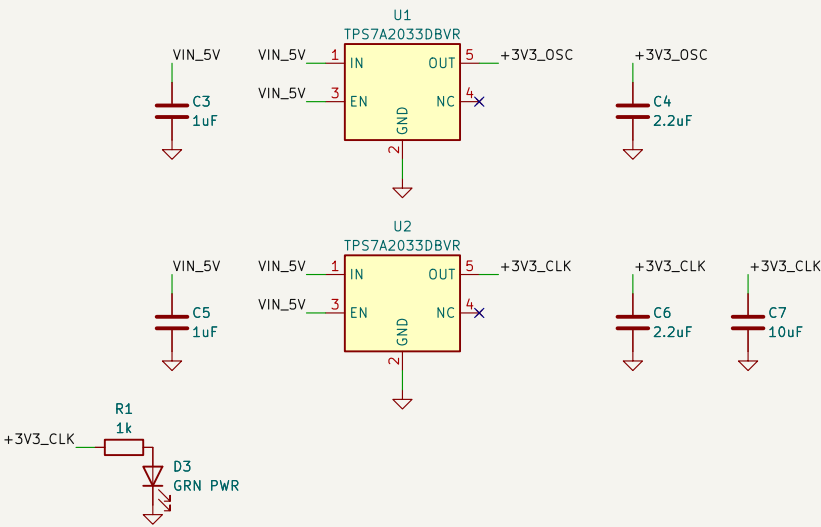


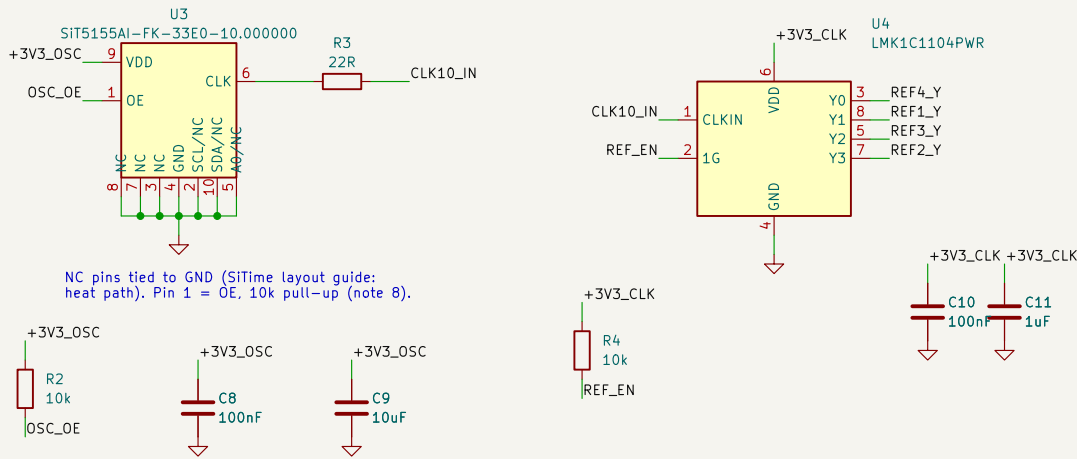
POWER INPUT: 5 V from payload rail (JST–GH, locking)



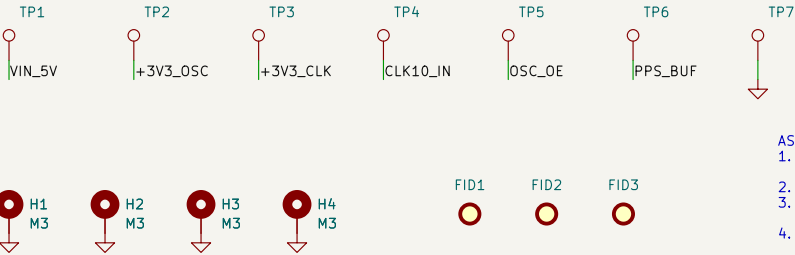
LOW-NOISE LDOs: separate rail for the TCXO



10 MHz SOURCE: SiT5155 Super–TCXO + LMK1C1104 1:4 buffer



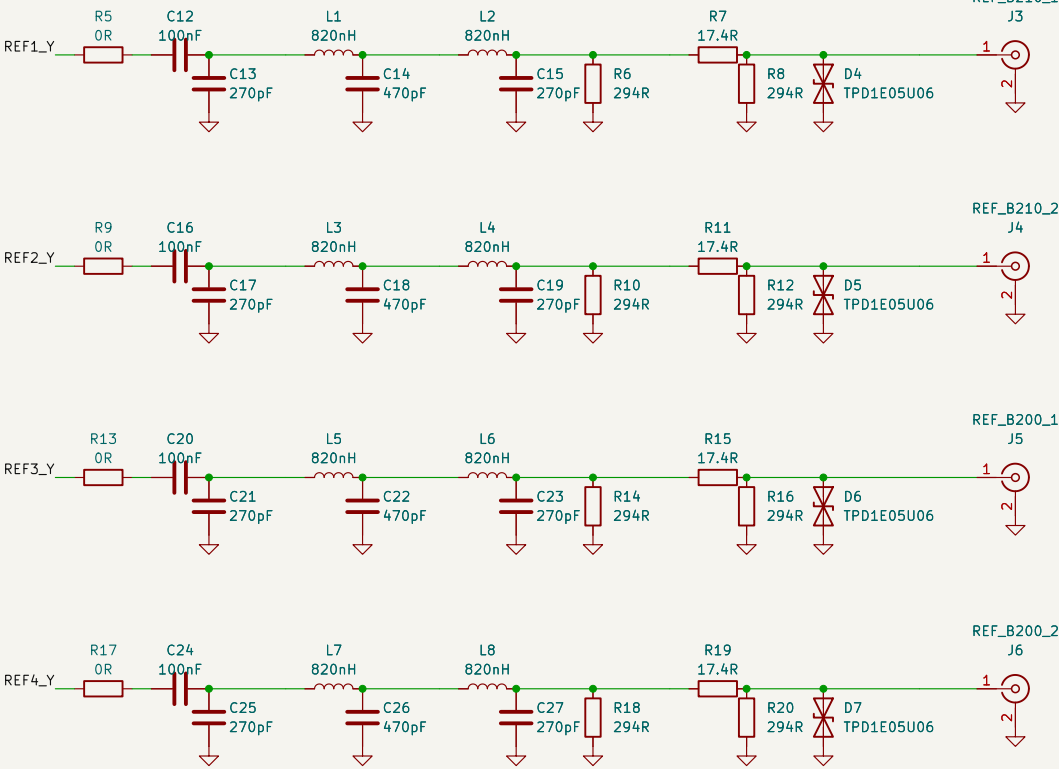
TEST POINTS / MECHANICAL



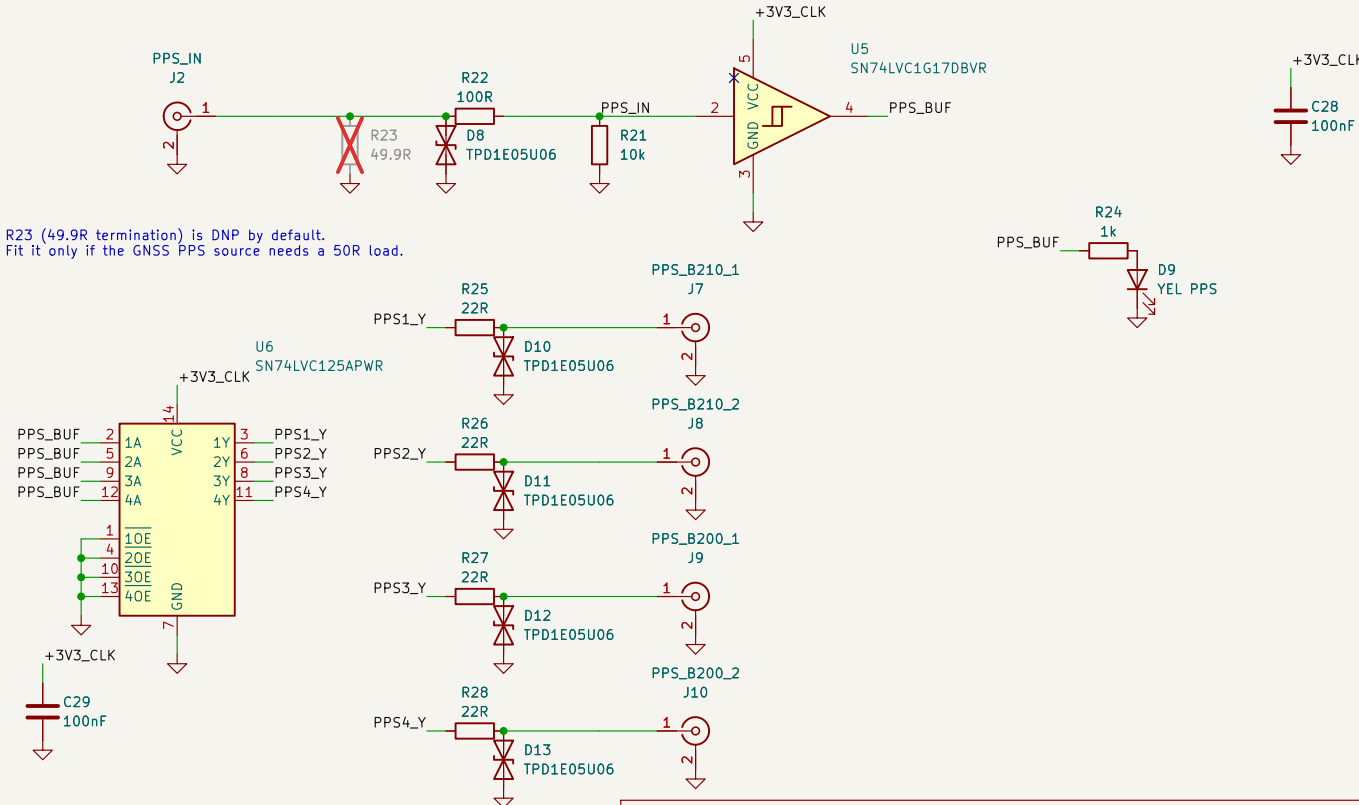
ASSEMBLY NOTES

1. U3 SiT5155: water-soluble flux only. No no-clean flux. No ultrasonic or megasonic cleaning (SiTime).
2. Reflow: IPC/JEDEC J-STD-020 profile.
3. J2 SMA (132134): hand-solder after reflow. J3–J10 SMP (SMP–MSLD–PCT): insert loose, mate the SDRs, then solder from the back (self-align).
4. Outputs: J3–J6 10 MHz ~ +7 dBm/50R, J7–J10 PPS 3.3 V CMOS.

10 MHz OUTPUTS x4: DC block -> 5th-order Chebyshev LPF (fc 13 MHz, 50R) -> 3 dB pad -> SMP



PPS: SMA in -> Schmitt buffer -> quad buffer -> 22R -> SMP x4 (3.3 V CMOS, DC coupled)



Generated by design.py – edit values there and re-run
SiT5155 Super–TCXO, 4x 10 MHz + 4x PPS to 2x B210 + 2x B200

CU Boulder – HERON

Sheet: /
File: heron_clock.kicad_sch

Title: HERON 10 MHz + PPS Distribution

Size: A3 Date: 2026–10–02

KiCad E.D.A. 10.0.6

Rev: B

Id: 1/1